

Optimization - 3

Gradient descent in action

Agenda

Recap

Intro to multivariable calculus

Partial derivatives

Intro to gradients

Representⁿ of gradients

Geometric interpretatⁿ of gradient

Computing gradient of any funcⁿ

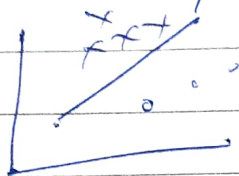
Computing partial derivative code

Intuitⁿ of gradient descent

Generalizatiⁿ of gradient descent.

maths

linear algebra

line $\rightarrow \mathbb{R}^m$

vector

 $\left\{ \begin{array}{l} \rightarrow \text{norm} \\ \rightarrow \text{angle} \end{array} \right.$

dot product

unit vector

project:

weight $\perp$ to line vector $\bar{x}$

$$\text{distance from origin} = \frac{-w_0}{\|w\|}$$

distance from point x

$$d = \frac{w^T x + w_0}{\|w\|}$$

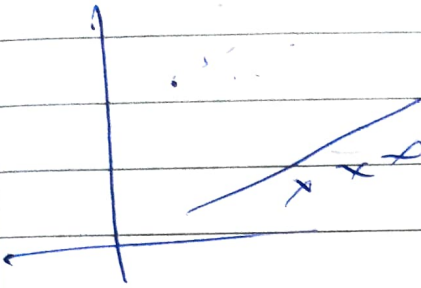
correcting the classical

$$w_{\text{new}} = w + \epsilon$$

Gain function

$$G = \max \left[\left(\frac{w^T x_i + w_0}{\|w\|} \right) \cdot y_i \right]$$

$$G = -1055.$$



$\Rightarrow$ good line but
not best classif~

change ω in range
-10 ... 10

find best line

This will take lots of time

To solve this
optimization problem

$$f(x) = 2x^2 + (x-3)^2 + (x-4)^2$$

$$f'(x) = 4x + 2(x-3) + 2(x-4)$$

$$f''(x) = 4 + 2 + 2 \\ = 8$$

→ this slope if $f'(x)=0$ then at x it will max or min

$$f'(x)=0 = 4x + 2x - 6 + 2x - 8 \\ = 8x - 14$$

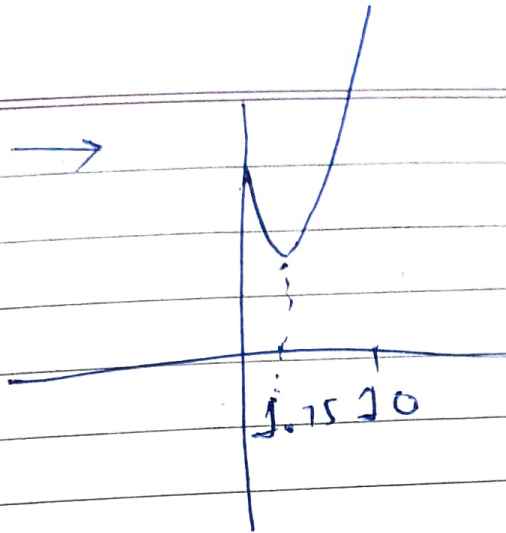
$$x = \frac{14}{8} = 1.75$$

is this max or min?

Need to find $f''(x)$

$$f''(x) = 8 > 0$$

so $x=1.75$ we will have
~~maxima~~ at $f(x)$
minima

$f(x) \rightarrow$ 

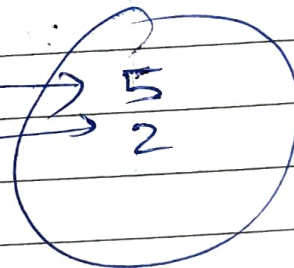
multivariable

$$f(x, y) = x^2 + y^2$$

Domain



Range



single value o/p
multiple value i/p

dependant
↓

↑ independent

$$y = x^2 + y^2 \rightarrow \text{independent}$$

$$y = mx + c$$

↑

↑

independent

dependant

$$y = m_1x_1 + m_2x_2 + m_3x_3 + c$$

↓

dependant

independent

y	x_1	x_2	x_3

all features x_1, x_2, x_3 should be independent

$$f(x, y) = x^2 + y^2$$

$$\frac{d}{dx} f(x, y) = \text{i want see change in } f \text{ w.r.t. } x$$

partial derivative

keeping y constant

$$= 2x$$

$$\frac{d}{dy} f(x, y) = 2y$$

$d = do, doh$

$$f(w_1, w_2, w_0) = w_1 x_1 + w_2 x_2 + w_0$$

$$\frac{d}{dw_1} = x_1$$

$$\frac{d}{dw_2} = x_2$$

$$\frac{d}{dw_0} = 1$$

↓ Vectors of derivative (slopes)

classmate

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$$\nabla f(\mathbf{x}) = \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

this is gradient

Q why taking derivative against ω instead of x

A →

$$f = \omega_1 x_1 + \omega_2 x_2 + \omega_0$$

here x_1, x_2 are fixed

these are data point.

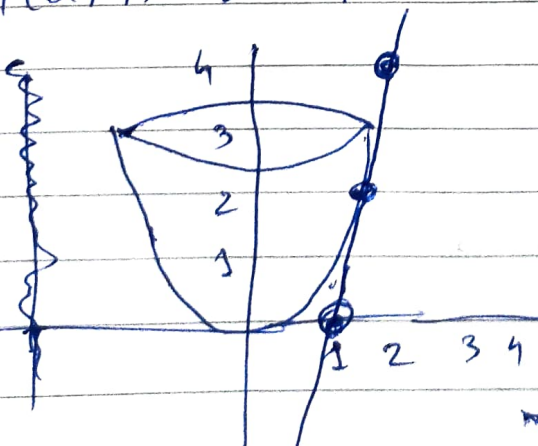
we ~~can~~ can't change

we can change ω to fit ~~curve~~ line

$$f(x, y) = x^2 + y^2$$

$$f'(x, y) =$$

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$



$\nabla f(x, y)$ at $x=2$.

$y=1$

$$\nabla f(2, 1) = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

single variable $\rightarrow$ derivative / slope

multiple variable $\rightarrow$ gradient

$$\frac{d}{dx} f(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{d}{dx} f(x, y) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\frac{d}{dy} f(x, y) = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Gradient descent



derivative



means slowly moving towards lower place

functⁿ $\rightarrow x^2$

goal $\rightarrow$ value of x so that
I get min value at x^2
here $x=0$

~~The~~

Goal of gradient descent

The entire goal is to find at what value the given functⁿ is minimum.

Now we will find min value of x^2 with gradient descent

① select random value

let say $x=7$

$$x^2 = 49$$

② decide directⁿ i.e. should I move < 7 or > 7

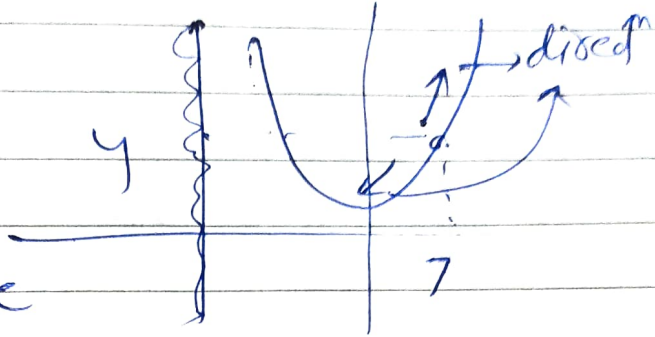
slope will give me the direct
slope = derivative = gradient

$$\frac{d}{dx} x^2 = 2x$$

$$= 14$$

$$> 0$$

+ve slope



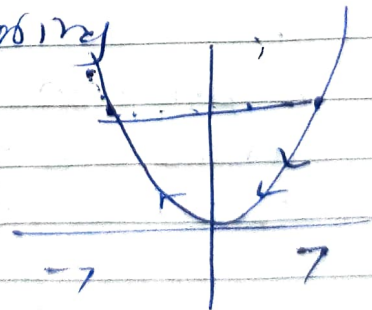
+ve slope means y is increasing
at $x=7$, we need to choose
downward direct.

As $\oplus$ slope, chose $\downarrow$ dirⁿ hence
subtract $d/dx x^2$ from original
chosen value i.e. 7

$$7 - 14 = -7$$

So next value at $x = -7$

Now $\frac{d}{dx} x^2 = -14$ -ve slope
means y is decreasing
~~here we missed~~



As we can from
figure see
we missed min
point $x=0$ so we need
to move lower

so we need

to move lower

So choose small step
 $x = 7 - \text{stepsize} \times 14$

$$\begin{aligned}x &= 7 - 0.1 \times 14 \\&= 7 - 1.4 \\&= 5.6\end{aligned}$$

$$x^2 = 31.36$$

we decreased x^2 from 49 to 31.36

$$\begin{aligned}\frac{d}{dx} x^2 &= 2x = 2 \times 5.6 \\&= 11.2\end{aligned}$$

Now new x

$$\begin{aligned}x &= 5.6 - 0.1 \times 11.2 \\&= 5.6 - 1.12 \\&= \end{aligned}$$

This is gradient descent algorithm by repeating above algorithm we will reach min point.

Gradient descent steps

① Make initial guess x' for $x(x')$ where $f(x)$ is considered min.

② Find slope at that point
if $f(x)$ at x' is +ve
then update x

$$x = x' - \text{stepsize} \cdot f'(x')$$

if $f(x)$ at x' is -ve

then update x

$$x = x' + \text{stepsize} \cdot f'(x')$$

③ Update guess using
GD rule

$$x = x' - \frac{df}{dx} \times \text{stepsize}$$



learning
rate

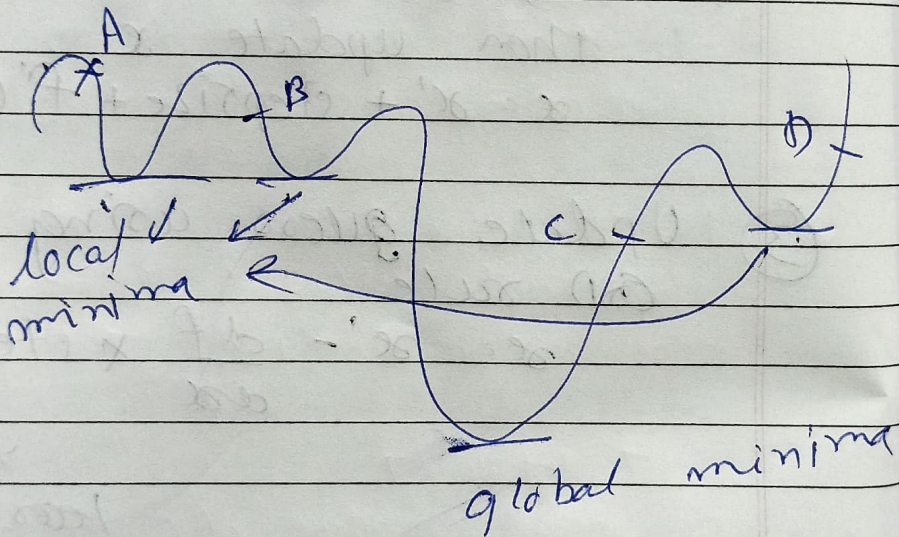
it is called
hyper
parameter

$$x^{(1)} = x^{(0)} - \eta \frac{d}{dx} f(x)$$

Gradient descent update
rule.

How to know reached min
~~at min~~ around min. x
i.e. $x \pm \Delta$ the slope will
be ~~constant~~ ≈ 0

If there are more than 1
minima.



When we start point A then
we will be stuck at local
minima, to avoid it we
start with multiple random
variable A, B, C, D
then we can find global
minima.